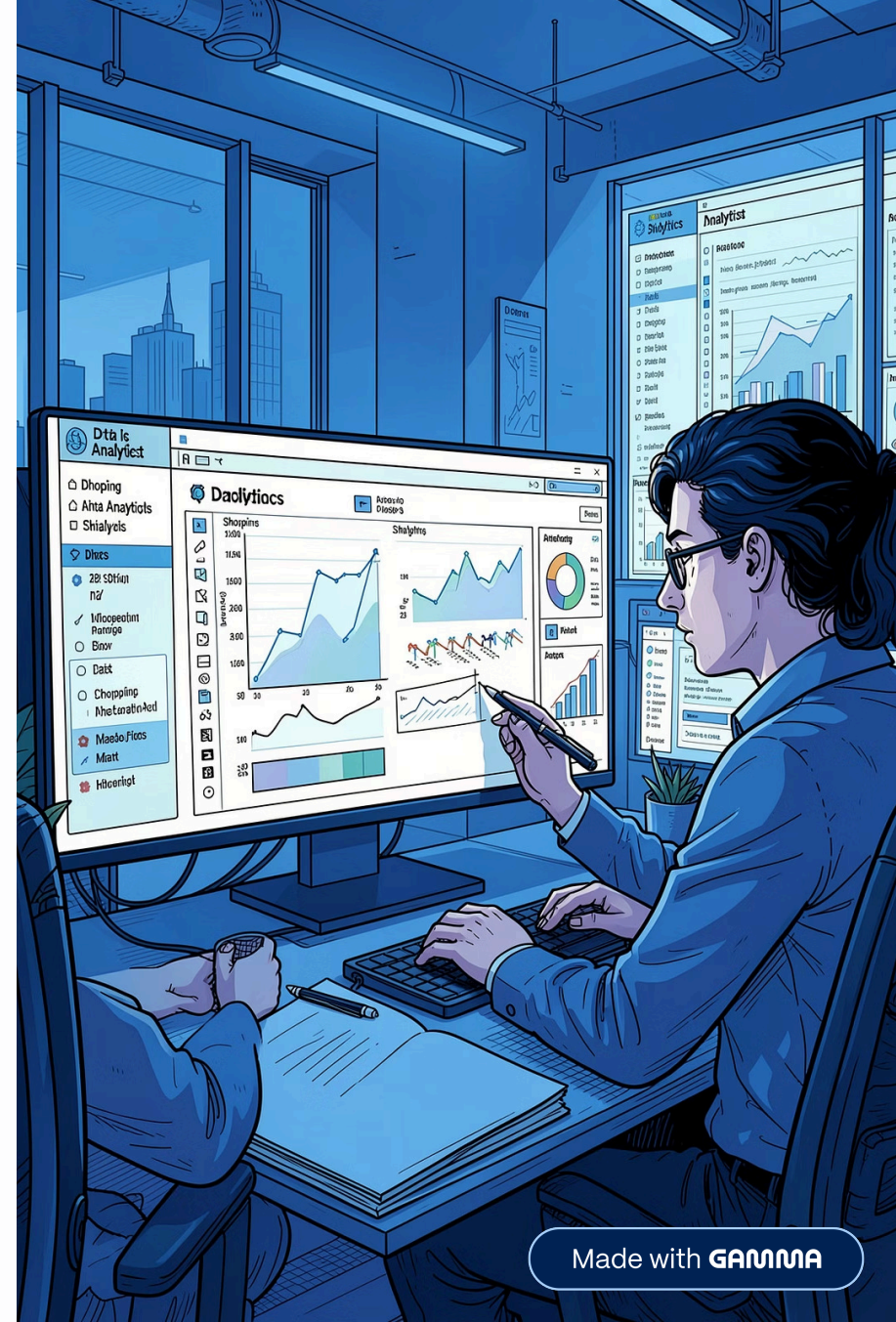


Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI



Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg. Purchase

Customer Data

Age, Gender, Location, Subscription Status

Behavior Data

Discounts, Promo Codes, Frequency, Review Ratings, Shipping

Data Preparation in Python

01

Load & Explore

`pandas · df.info() · describe()`

02

Clean

Median imputation for Review Rating · snake_case columns

03

Engineer

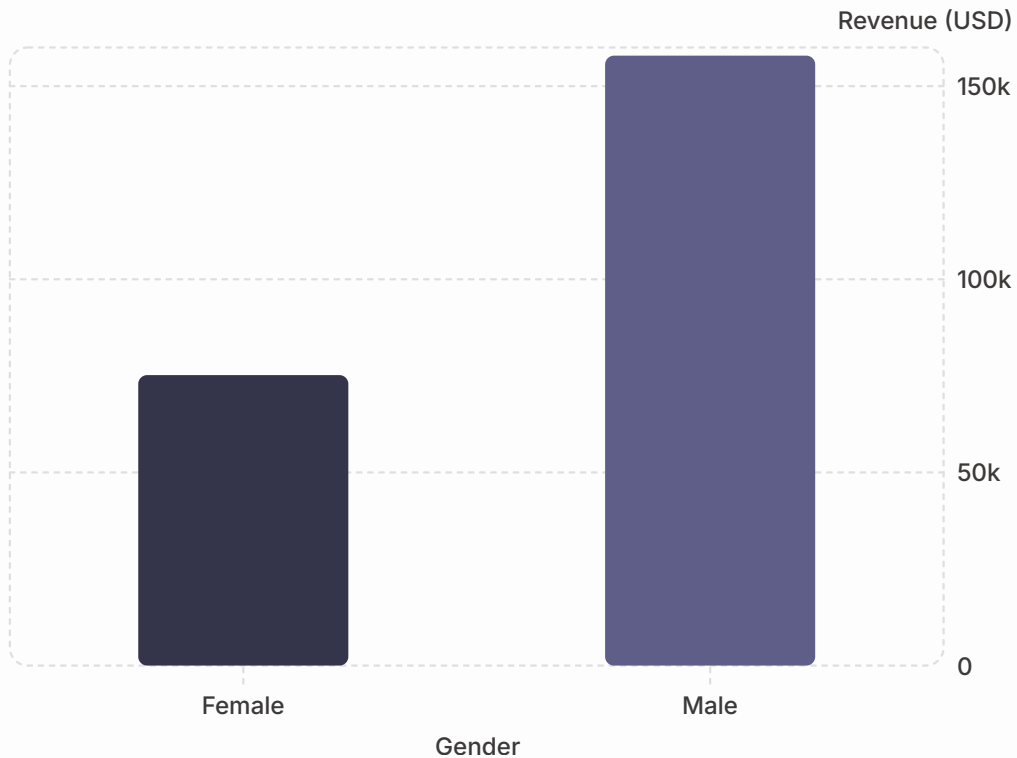
`age_group bins · purchase_frequency_days`

04

Integrate

Loaded cleaned DataFrame into PostgreSQL

Revenue by Gender



Key Insight

Male customers generated **2.1× more revenue** than female customers.

i Male: \$157,890 · Female: \$75,191

Top Products & Discount Patterns

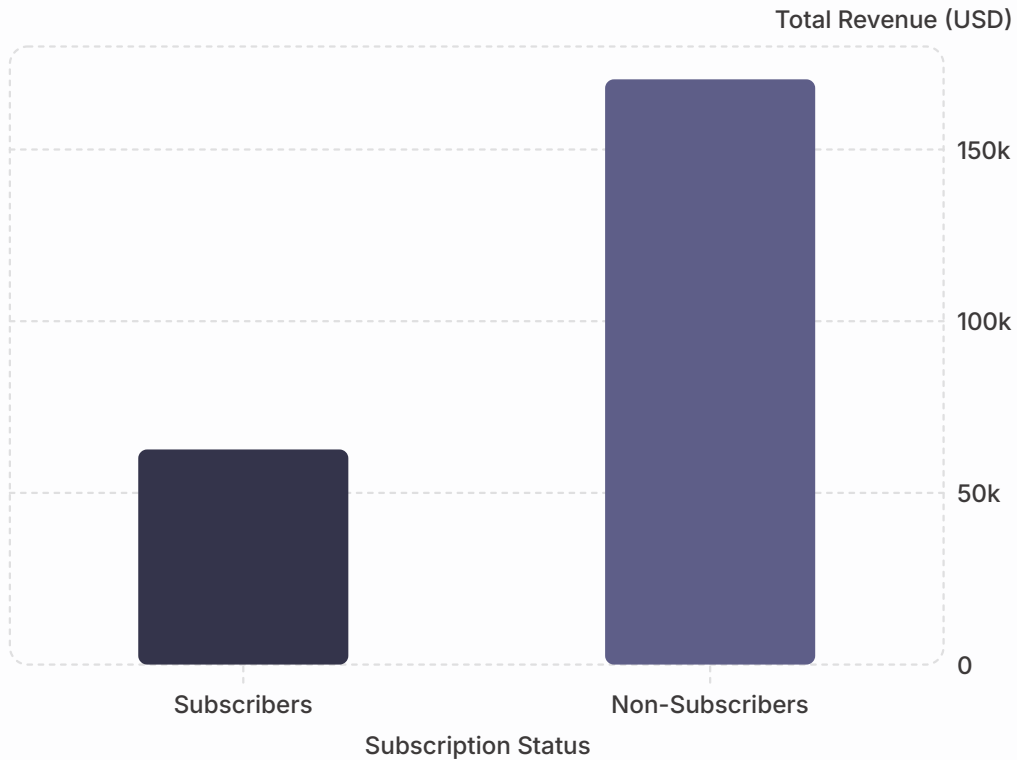
Top 5 by Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Product	Discount Rate
Hat	50%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Subscribers vs. Non-Subscribers



Comparison

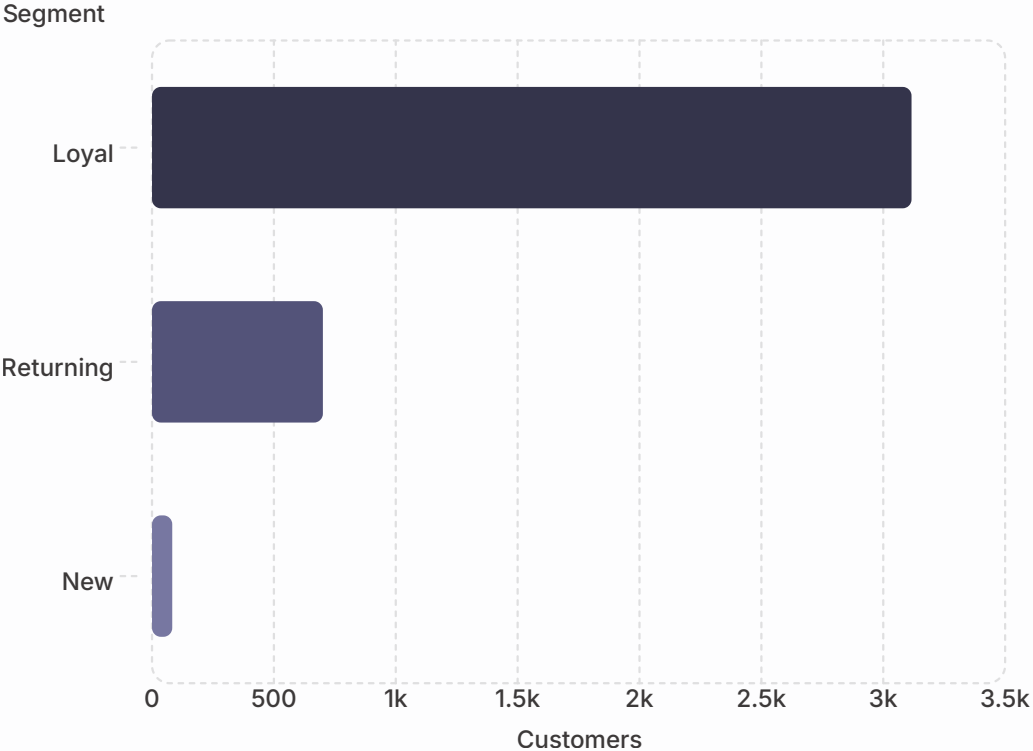
Subscribers (27%)

1,053 customers ·
\$59.49 avg · \$62,645
total

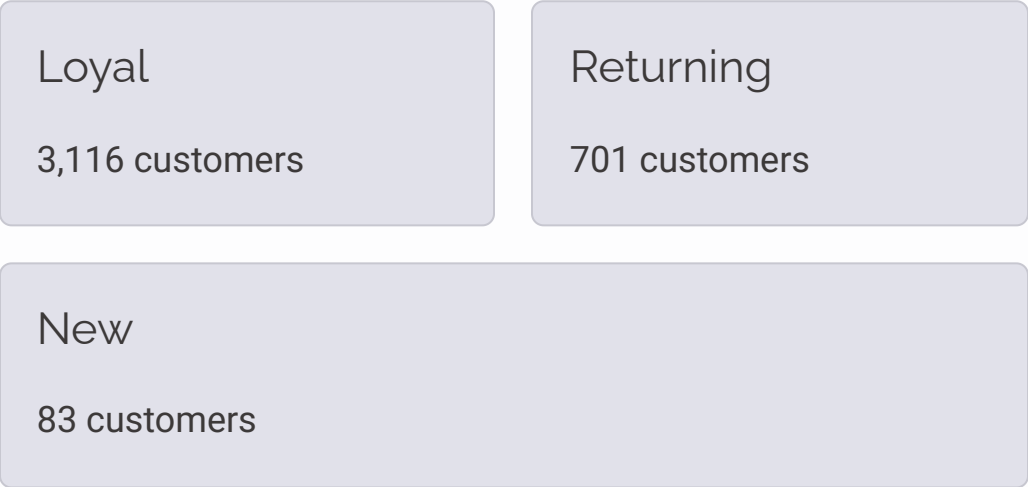
Non-Subscribers
(73%)

2,847 customers ·
\$59.87 avg · \$170,436
total

Customer Segmentation

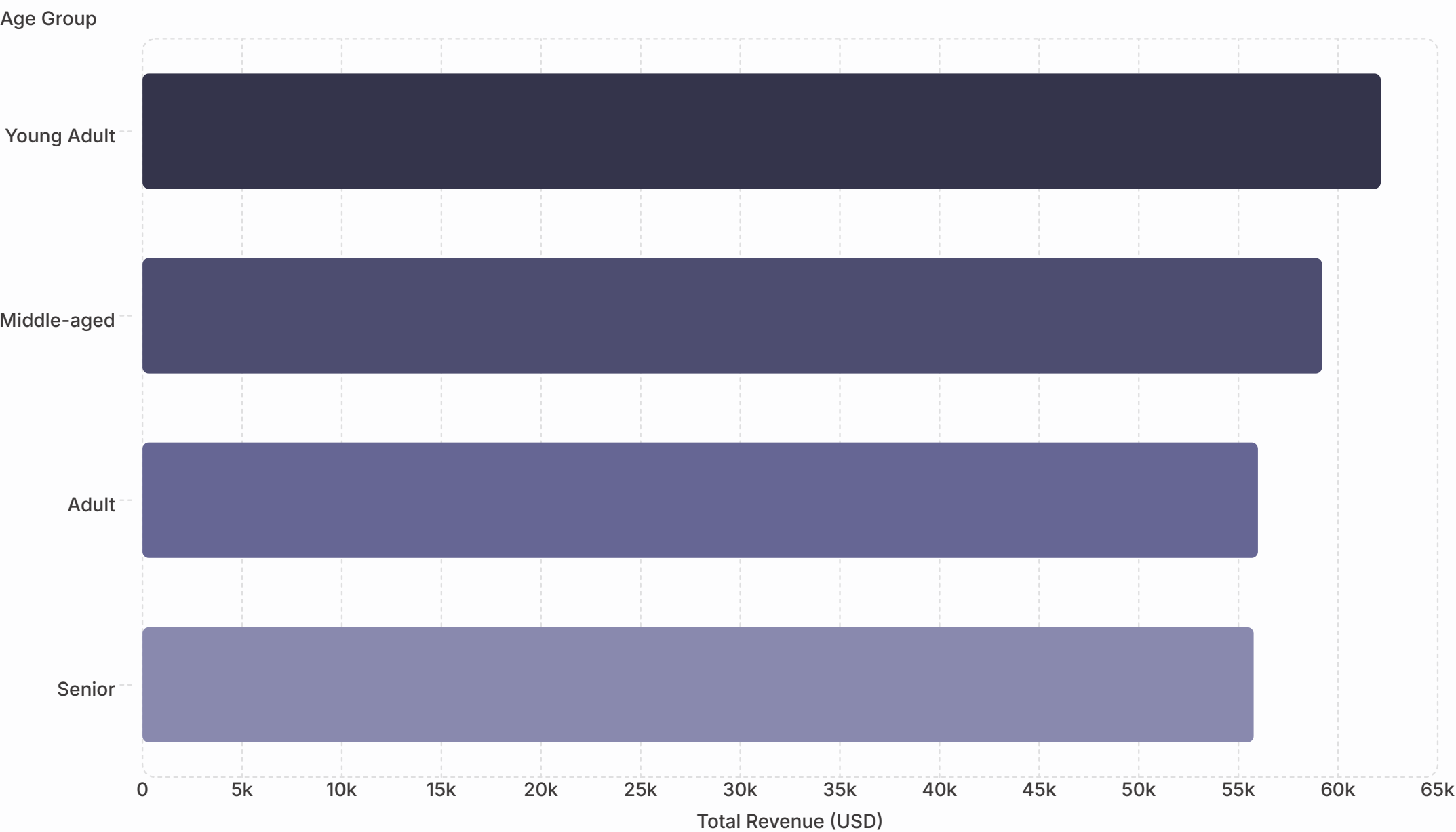


Segment Breakdown



Repeat buyers: 2,518 unsubscribed vs. 958 subscribed

Revenue by Age Group



Young Adults lead revenue contribution, though all four age groups are relatively close in total spend.

Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription · Gender · Category
· Shipping Type

Visuals

Revenue & Sales by Category · Revenue & Sales by Age Group ·
Subscription %





Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users